# 3DPS: Differentiable 3D Point Splatting for Physically Based mmWave Radar Rendering and Novel View Synthesis

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## Abstract

Achieving high-fidelity novel view synthesis (NVS) for millimeter-wave (mmWave) radar requires a renderer that is multi-viewpoint-tractable, physically faithful, and complex-valued. No prior method has all three. Differentiable Monte Carlo (MC) ray tracers implement the radar forward model directly with explicit material modeling and complex outputs, but do not scale to the multi-view optimization NVS demands. Optical-NVS ports of NeRF, hash grids, and 3D Gaussians train fast but trade explicit modeling of the radar signal chain for runtime, supervising on post-FFT power-only range–azimuth (RA) magnitudes with phase dropped and the radar's signal-model components absorbed into learned features. We derive 3D Point Splatting (3DPS), the first differentiable point renderer for radar, directly from the standard solid-angle form of the radar equation. Each oriented 3D point carries an ITU-R P.2040 material model, evaluated in closed form, with the resulting complex phasor splatted into range bins through a precomputed point spread function (PSF). Complex-valued output is product-agnostic: the same optimized scene synthesizes raw analog-to-digital converter (ADC) samples, complex range profile (CRP), and RA outputs at novel viewpoints through standard FFT pipelines without retraining for each format. On six outdoor ColoRadar scenes, 3DPS reaches **0.587** mean Pearson correlation on held-out RA images, $1.8$–$7.2\times$ over RadarSplat, Radar Fields, and DART. On CRP and ADC outputs, which no baseline produces, 3DPS reaches **0.603** and **0.613** mean Pearson correlation respectively. Training takes $\sim$3 minutes per scene on a single RTX 4090, $\sim$50$\times$ faster than the SOTA MC ray tracer's projected per-scene cost.

## 1 Introduction

Millimeter-wave (mmWave) radar penetrates fog, rain, and airborne dust Guan et al. [2020], returns Doppler velocity directly, and is an essential sensor for robust perception in autonomous driving and robotics Gao et al. [2019]. However, radar datasets are substantially smaller than their optical or LiDAR counterparts because radar sensors are not ubiquitous, the captured data structure depends heavily on array geometry and application-specific waveform configuration, and most large-scale datasets remain proprietary. NVS for radar can address this data scarcity by enabling sparse training sets to generalize across novel poses, but this problem is structurally hard. The supervision signal is typically i) sparse with returns dominated by quasi-mirror specular reflections at 60–77 GHz, ii) constrained to 2D as COTS radar array geometries lack an elevation aperture and integrate out the elevation dimension, and iii) limited in view count as NVS aggregates only a handful of training views per scene.

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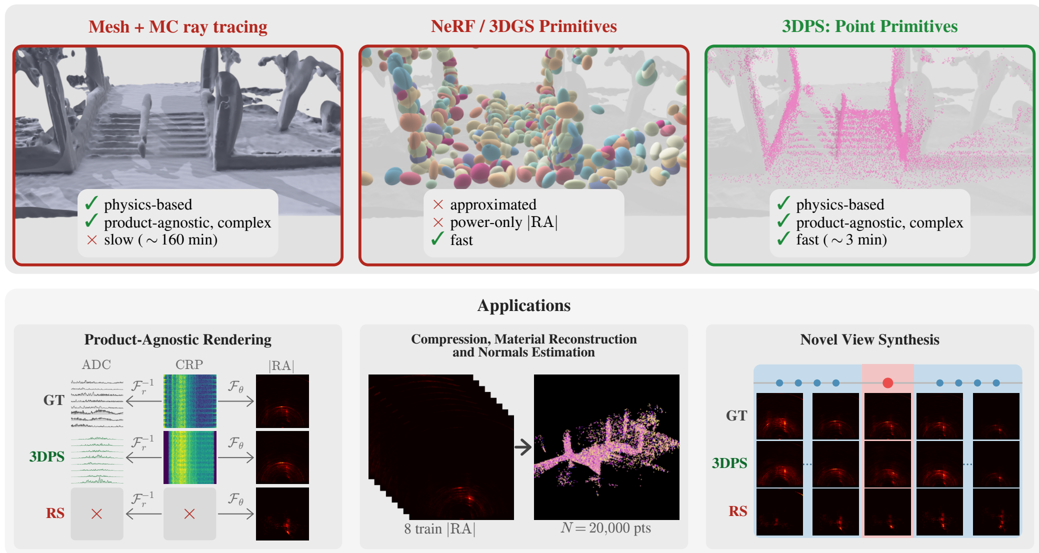


Figure 1: **The right primitive for radar NVS.** Top: three method families on the three properties radar NVS demands (multi-viewpoint-tractable, physics-faithful, complex-valued). MC ray-tracing renderers (mmIR, Sionna RT) are physically faithful but too slow for multi-viewpoint optimization; optical-NVS adaptations (RadarSplat, Radar Fields, DART) are fast but discard phase. 3DPS is the first to satisfy all three. Bottom: 3DPS outputs on the held-out test frame of S1 F185 — $|\mathrm{RA}|$ render, raw ADC magnitude (inverse range FFT), and the optimized point cloud colored by learned $\varepsilon_r'$.

Achieving high-fidelity radar NVS requires a renderer with three properties. First, multi-viewpoint-tractable, since NVS by definition demands joint optimization across many training poses. Second, physically faithful, so that predictions at unseen poses match the underlying sensor physics and the optimized scene parameters remain interpretable, editable, and transferable across sensor views. Third, complex-valued, since the radar forward model emits a complex phasor that is product-agnostic: the same optimized scene synthesizes raw analog-to-digital converter (ADC) samples, complex range profile (CRP), and range–azimuth (RA) outputs at novel viewpoints through standard FFT pipelines, without retraining (Fig. 1). We show that this also allows the renderer to reproduce antenna patterns, sidelobes, and FFT windowing from the signal model, sharpening RA correctness. Phase-discarding renderers forfeit both.

No prior method has all three properties. Differentiable Monte Carlo (MC) ray tracers Hoydis et al. [2023], Hofmann et al. [2025], Chen et al. [2025], Anonymous [2026] implement the radar forward model with explicit BSDF material modeling, real antenna beam patterns, and complex outputs. The cost is structural, since each surface hit must be evaluated against every (TX, RX) pair to maintain MIMO coherence. The closest prior work, mmIR Anonymous [2026], requires ~20 minutes per single-viewpoint fit at 24k surface interactions, which does not scale to the multi-view optimization NVS demands. Optical-NVS adaptations Huang et al. [2024], Borts et al. [2024], Takawale et al. [2025], Kung et al. [2025] of neural radiance fields, hash grids, and 3D Gaussians Mildenhall et al. [2020], Kerbl et al. [2023] train fast on real radar captures, because they trade explicit modeling of the radar signal chain for runtime. Supervision is on post-FFT power-only RA images, phase is dropped by construction, and the radar's signal-model components are absorbed into learned features rather than rendered.

3DPS realizes all three properties, because we derive it directly from the standard solid-angle form of the radar equation Skolnik [2001], Anonymous [2026]. For multi-viewpoint tractability, we splat each point's complex phasor into range bins through a precomputed Hann-windowed point spread function (PSF) at constant cost per (point, range bin), fuse BSDF evaluation and splatting into three CUDA kernels, and exploit a quasi-static MIMO factorization that linearizes the BSDF cost in $N_{\mathrm{TX}} + N_{\mathrm{RX}}$. For physical fidelity, the antenna geometry, BSDF (per ITU-R P.2040), and antenna gains are evaluated in closed form with no learned features, and phase is integrated from the geometric path length rather than learned. For complex-valued output, the per-(TX, RX) range

Table 1: Comparison of radar rendering methods. Signal types: ADC, CIR (channel impulse response), IF (intermediate-frequency phasors), RA, RDA (range–Doppler–azimuth cube). Repr. = scene representation. Mat. = explicit material model. Per-pt = per-point/per-vertex. Real = validated on real data. Time = per-scene wall-clock on RTX 4090 over 8 training viewpoints (Sec. 4).

| **Method** | **Repr.** | **Signal** | **Complex** | **Diff.** | **Mat.** | **Per-pt** | **BSDF** | **NVS** | **Real** | **Time** |
|---|---|---|---|---|---|---|---|---|---|---|
| Sionna RT Hoydis et al. [2023] | Mesh | CIR | ✓ | Partial | – | – | – | – | – | – |
| SFCW Inv. Hofmann et al. [2025] | Mesh | IF | ✓ | Partial | ✓ | ✓ | – | – | ✓ | – |
| InverTwin Chen et al. [2025] | Mesh | CIR | ✓ | ✓ | – | – | – | – | ✓ | – |
| mmIR Anonymous [2026] | Mesh | ADC | ✓ | ✓ | ✓ | ✓ | ✓ | – | ✓ | 160 min |
| DART Huang et al. [2024] | Implicit | RDA | – | ✓ | – | – | – | ✓ | ✓ | 0.4 min |
| Radar Fields Borts et al. [2024] | Implicit | RA | – | ✓ | – | – | – | ✓ | ✓ | 1.6 min |
| SpINR Takawale et al. [2025] | Implicit | RA | – | ✓ | – | – | – | ✓ | ✓ | – |
| RadarSplat Kung et al. [2025] | 3DGS | RA | – | ✓ | – | – | – | ✓ | ✓ | 20 min |
| **Ours** | Points | ADC, CRP, RA | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 3 min |

profile is preserved end-to-end through PSF splatting, enabling product-agnostic rendering to ADC, CRP, and RA without retraining via standard FFT pipelines.

We evaluate sparse-view radar NVS on six outdoor ColoRadar Kramer et al. [2022] scenes, measuring all three downstream products from the same optimized scene. On the held-out test view, 3DPS reaches **0.587** mean Pearson correlation on $|\mathrm{RA}|$, beating RadarSplat $(0.333)$, Radar Fields $(0.120)$, and DART $(0.081)$ by $1.8\times$, $4.9\times$, and $7.2\times$ respectively. Further, 3DPS reaches **0.603** mean correlation on $|\mathrm{CRP}|$ and **0.613** on the ADC envelope. Training takes $\sim$3 minutes per scene on a single RTX 4090 across 8 training viewpoints jointly, $\sim 50\times$ faster than the SOTA MC ray tracer's projected $\sim 160$ min per-scene cost.

**Contributions.** **(i)** The first differentiable point renderer for radar, with a radar-native primitive (oriented 3D points carrying ITU-R P.2040 material parameters) and multi-viewpoint-tractable rendering operation (PSF splatting of complex phasors into range bins), derived from the solid-angle form of the radar equation. **(ii)** Product-agnostic rendering, since the same optimized scene yields CRP, ADC, and RA outputs through standard FFT pipelines without retraining. **(iii)** Sparse-view radar NVS at the state of the art on $|\mathrm{RA}|$, $|\mathrm{CRP}|$, and ADC envelope across six ColoRadar scenes, $\sim 50\times$ faster per scene than the SOTA MC ray tracer.

## 2 Related Works

**Physics-based inverse rendering for radar.** Sionna RT Hoydis et al. [2023] differentiates antenna patterns and scattering coefficients but suspends gradients through its path solver and does not model wideband FMCW. Hofmann et al. Hofmann et al. [2025] estimate per-vertex stepped-frequency materials but assume known geometry and single-bounce propagation. InverTwin Chen et al. [2025] addresses multipath gradient pathologies via path-space differentiation, but uses a Gaussian-kernel surrogate for range processing and validates on synthetic data only. mmIR Anonymous [2026] is the closest prior work and the only fair MC comparison: a differentiable ray tracer with per-vertex ITU materials, normals, and antenna patterns on real captures, but its $\sim$20 min per-viewpoint cost makes the multi-view optimization NVS demands intractable.

**Neural fields for radar novel view synthesis.** DART Huang et al. [2024] trains a NeRF on range–Doppler magnitude. Radar Fields Borts et al. [2024] synthesizes power-only RA via hash-grid + MLP. SpINR Takawale et al. [2025] models FMCW occupancy implicitly. RF-4D Zhang et al. [2025], NeuRadar Rafidashti et al. [2025], and NeWRF Lu et al. [2024] extend to dynamic scenes or fuse radar with RGB/LiDAR but render no complex output.

**Gaussian splatting for radar.** 3DGS Kerbl et al. [2023] rasterizes anisotropic 3D Gaussians for real-time optical NVS. RadarSplat Kung et al. [2025] ports this primitive to RA rendering with opaque per-Gaussian features, providing no explicit material model and no complex output. The mismatch is structural: 3D Gaussians were designed for optical radiance fields where each primitive emits view-dependent RGB via spherical harmonics, an output space that does not naturally parameterize complex phasors at TX–RX-specific carrier phase and amplitude. The 3DGS limitations on point-based primitives were calibrated to sub-pixel optical resolution: point splats "suffer from holes, causing aliasing" and are "strictly discontinuous" Kerbl et al. [2023]. None of these bind at radar resolution. The cascade resolves at $\Delta r$=5.93 cm in range and $\sim 1.4°$ in azimuth, and each oriented point splats a continuous Hann-windowed PSF across fifteen range bins ($\sim$89 cm) and multiple azimuth bins after the virtual-array FFT. Conversely, the MC ray tracers we compete against (mmIR Anonymous [2026], Sionna RT Hoydis et al. [2023]) already compute scene hit points to synthesize ADC; promoting those hits to persistent learnable primitives eliminates the per-iteration ray-tracing cost while preserving the physical structure that makes hit-based rendering correct.

# 3 Method

3DPS renders complex-valued FMCW range profiles from a learned set of *oriented 3D points* anchored on a LiDAR-derived geometric scaffold. Each point carries a quaternion-encoded surface normal and six physically interpretable ITU-R P.2040 material parameters, evaluated by a closed-form bidirectional scattering distribution function (BSDF) and splatted into range-profile bins through a precomputed sensor point spread function (PSF). The pipeline is fully deterministic: there is no Monte Carlo sampling, no learned antenna pattern, and no random-direction integration. Sec. 3.1 states the radar forward model. Sec. 3.2 presents the deterministic point-splatting renderer, our central contribution, and the LiDAR-derived geometric prior. Sec. 3.3–3.4 specify the per-point ITU BSDF and the differentiable optimization.

## 3.1 Radar forward model

For a single TX–RX pair observing a point target at $\mathbf{x}$ with normal $\mathbf{n}$, the bistatic radar equation Skolnik [2001] relates received power to bistatic radar cross-section, transmit power $P_t$, antenna gains $G_t, G_r$, wavelength $\lambda$, and TX/RX path lengths $d_t, d_r$. Replacing the point-target RCS with a spatially varying BSDF $f(\boldsymbol{\omega}_t, \mathbf{x}, \boldsymbol{\omega}_r)$ and integrating over the receiver hemisphere $\Omega_{rx}$ Anonymous [2026],

$$P_r \;=\; \frac{P_t\,\lambda^2}{(4\pi)^2}\int_{\Omega_{rx}} \frac{G_t(\boldsymbol{\omega}_t)\,G_r(\boldsymbol{\omega}_r)\,f(\boldsymbol{\omega}_t,\mathbf{x},\boldsymbol{\omega}_r)\,c_t\,V}{d_t^2}\,d\omega_r, \tag{1}$$

where $c_t = \max(0, \mathbf{n}\cdot\boldsymbol{\omega}_t)$ is the cosine of incidence and $V \in \{0,1\}$ is binary visibility. Standard FMCW renderers approximate Eq. (1) via Monte Carlo path sampling, synthesize ADC samples per path, and apply a range FFT to recover range bins; mmIR Anonymous [2026] is the current state of the art for this pipeline and requires $\sim$20 minutes for a 500-iteration single-viewpoint fit at 24k surface interactions per iteration.

## 3.2 Deterministic point splatting

**Deterministic evaluation.** 3DPS evaluates the forward model deterministically: each oriented point in the scene contributes *exactly once* per (TX, RX) pair, with no Monte Carlo path sampling and no random-direction integration. The hemisphere integral of Eq. (1) is replaced by a closed-form sum over the fixed point set $\{\mathbf{x}_i\}_{i=1}^{N}$ with $N$=20,000, evaluated at the deterministic per-pair incidence and observation angles obtained directly from the array geometry.

**Range-profile splatting.** The range FFT of a single-path ADC contribution is a Hann-windowed sinc pulse centered at the path's bistatic range bin, fully determined by the chirp bandwidth and the FFT window. We precompute this point-spread function $\mathrm{PSF}(r) \in \mathbb{C}^K$ once on a $K$=256 bin grid at the radar's range resolution $\Delta r = c/(2\,B_{\text{chirp}}) = 5.93$ cm, where $B_{\text{chirp}}$ is the chirp's swept bandwidth. The forward pass then becomes a sparse scatter-add directly into range-profile bins:

$$\mathrm{RP}_{TR}[k] \;=\; \sum_{i=1}^{N} A_i^{TR}\,\exp(-j\,\phi_{TRi})\,\mathrm{PSF}\big(k - k^\star_{TRi}\big), \quad k^\star_{TRi} = \mathrm{round}(r_{TRi}/\Delta r), \tag{2}$$

where $A_i^{TR} \in \mathbb{C}$ collects the antenna gains, BSDF, and geometric falloff factors of Eq. (1) for point $i$, $\phi_{TRi} = 2\pi(d_{Ti}+d_{Ri})/\lambda$ is the round-trip carrier phase, and $r_{TRi} = (d_{Ti}+d_{Ri})/2$ is the bistatic range. Splatting into the $L$=15 nearest range bins per path replaces the entire fast-time DFT, dropping the cost from $\mathcal{O}(N\cdot K)$ to $\mathcal{O}(N\cdot L)$ per (TX, RX) pair.

**Validity of the splat.** The Hann-windowed PSF tail beyond $L$=15 taps is below $-50$ dB relative to peak, well under the noise floor of real ColoRadar captures. The bin-rounding $k^\star = \mathrm{round}(r/\Delta r)$

introduces at most $\Delta r/2 = 2.97$ cm of range bias per path, an order of magnitude below the radar's intrinsic range resolution; the rounding is applied only to the splat index and does not affect the carrier phase $\phi$, which is computed from the un-rounded $d_T + d_R$. Sidelobes and FFT windowing artifacts are captured exactly by the precomputed PSF. The splat is therefore numerically equivalent to the standard ADC-then-FFT pipeline on the same path set: we validate this empirically against mmIR's MC reference at maximum forward error $1.19\times10^{-7}$ on 90k paths Anonymous [2026].

**MIMO factorization.** The cascade radar's $N_{\text{TX}}$=12 transmitters and $N_{\text{RX}}$=16 receivers produce 192 TX–RX pairs per chirp. The ITU BSDF (Sec. 3.3) factors over incidence and outgoing directions, so we evaluate the per-point incidence factor $f_i^{\text{in}}(\hat{\mathbf{d}}_{Ti})$ once per TX, the outgoing factor $f_i^{\text{out}}(\hat{\mathbf{d}}_{Ri})$ once per RX, and combine them per pair without recomputing material physics — dropping the BSDF cost from $\mathcal{O}(N\cdot N_{\text{TX}}\cdot N_{\text{RX}})$ to $\mathcal{O}(N\,(N_{\text{TX}}+N_{\text{RX}}))$, the ray-reuse strategy of mmIR Anonymous [2026], Schüßler et al. [2021]. After splatting, an $N_{\text{TX}}$-point virtual-array DFT along the azimuth dimension converts the per-(TX, RX) range profiles into a range–azimuth (RA) map of shape $(N_{\text{az}}, K) = (127, 256)$. The $N_{\text{az}}$=127 azimuth bins correspond to the unique TDM-MIMO virtual-element positions of the cascade array geometry, padded to the standard arcsin-spaced grid $\theta_a = \arcsin(2(a - N_{\text{az}}/2)/(N_{\text{az}}+1))$ with forward-cone half-angle $\arcsin(126/128) \approx 79.86°$.

**CUDA fusion.** The combined cost of Eq. (2) plus factorized BSDF plus virtual-array DFT is implemented as three fused CUDA kernels: a forward BSDF kernel evaluating the GGX, SPM, and diffuse lobes plus Jones-polarized Fresnel coefficients in fp32; a scatter-splat kernel performing Eq. (2) via atomic adds into range-profile bins; and an analytical backward kernel computing per-point partial derivatives via atomic accumulation, with a closed-form Jacobian for the slab-Fresnel coefficients. The analytical backward is validated against finite-difference gradients to within numerical tolerance. The fused pipeline renders $N$=20,000 scene points across all 192 (TX, RX) pairs in $\sim$10 ms on an RTX 4090, and a 500-iteration training pass over the 8 training viewpoints completes in $\sim$3 minutes per scene.

**LiDAR-derived geometric prior.** We initialize the point set $\{\mathbf{x}_i\}$ from a co-registered LiDAR scan. A 3D geometric prior is mandatory, not optional: the forward model in Eqs. (1)–(2) requires full 3D positions $\mathbf{x}_i \in \mathbb{R}^3$ to compute bistatic range $r_{TRi}$, incidence/observation angles $\boldsymbol{\omega}_t, \boldsymbol{\omega}_r$, and the elevation-dependent antenna gains $G_t G_r$ (the cascade's planar virtual array has a non-trivial elevation pattern even though it lacks an elevation aperture). The supervision signal — 2D range–azimuth maps — has integrated out the elevation dimension by construction, so RA maps alone are information-theoretically insufficient to recover 3D point positions regardless of how many viewpoints are aggregated. Compounding this, NVS is performed from only eight training views per scene, and each individual RA map is itself extremely sparse: mmWave returns are dominated by strong specular reflections at near-normal incidence with sparse glints elsewhere, since outdoor surfaces tend to act as quasi-mirrors at 60–77 GHz Guan et al. [2020], Lu et al. [2013]. LiDAR-conditioned scaffolds are accordingly the dominant precedent for radar rendering and ray-tracing on real scenes Dudek [2010], Schüßler et al. [2021], Mostajabi et al. [2020], Kramer et al. [2022], Anonymous [2026], Kung et al. [2025], Rafidashti et al. [2025], Hoydis et al. [2023]. We reduce the per-scene LiDAR cloud (positions, normals, intensity; $\sim$ 2.7–6.5M points across our scenes) to $N$=20,000 oriented points by (1) culling outside the radar's azimuth cone ($\cos(\angle(\hat{\mathbf{d}}, \hat{\mathbf{b}})) > 0.1761$) and radial range (1.5 m to $K\Delta r$=15.2 m); (2) Mitsuba 3 ray-casting Jakob et al. [2022] from each surviving point toward the array centroid to drop occluded points; (3) cosine-weighted resampling of $3N$ candidates with replacement at probability $\propto \max(\cos(\angle(\hat{\mathbf{d}}, \hat{\mathbf{b}})), 0.01)$, biasing selection toward boresight-aligned regions where the antenna pattern is strongest; and (4) farthest-point sampling to exactly $N$=20,000. Each surviving point inherits a quaternion encoding its LiDAR surface normal and a uniform ITU-concrete material prior.

### 3.3 Per-point ITU BSDF

Each point's BSDF follows the ITU-R P.2040 multilayer slab model used by mmIR Anonymous [2026], ITU-R [2023], Wei et al. [2024], applied per-point rather than per-vertex. The 6-parameter material vector $\boldsymbol{\theta}_i = (\varepsilon_r', \varepsilon_r'', \sigma_h, l_c, \tau, d)_i$ specifies the complex relative permittivity $\varepsilon_r = \varepsilon_r' - j\,\varepsilon_r''$, RMS surface roughness $\sigma_h$, correlation length $l_c$, Kirchhoff-approximation / small-perturbation (KA/SPM) blend $\tau \in [0, 1]$, and slab thickness $d$. Each parameter is reparameterized via sigmoid (KA/SPM blend) or shifted softplus (positive-only quantities) to keep optimization in physically valid ranges.

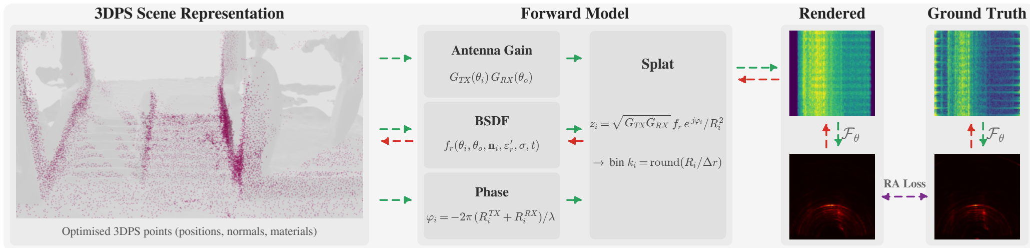


Figure 2: **3DPS rendering pipeline.** A LiDAR point cloud is reduced to $N$=20,000 oriented points (Sec. 3.2, LiDAR-derived geometric prior). For each (TX, RX) pair, the per-point ITU BSDF (Sec. 3.3) is evaluated at the deterministic incidence and observation angles, producing a complex amplitude $A_i^{TR}$ and a carrier phase $\phi_{TRi}$. Each point's complex contribution is splatted via a Hann-windowed PSF into the $L$=15 nearest range bins (Eq. 2). Per-(TX, RX) range profiles are stacked and an azimuth FFT is applied to obtain the final RA map.

The BSDF decomposes into coherent and incoherent components via the Rayleigh roughness factor $\eta = \exp(-(2k\sigma_h \cos\theta_t)^2)$ with $k = 2\pi/\lambda$:

$$f \;=\; A(\boldsymbol{\omega}_t)\,\big[\eta\, f_{\text{coh}} + (1-\eta)\, f_{\text{inc}}\big], \tag{3}$$

where $A(\boldsymbol{\omega}_t) \in [0, 1]$ is the slab-Fresnel energy gate (multilayer slab with thickness $d$ and complex permittivity $\varepsilon_r$, capturing the internal reflections that single-interface Fresnel misses — critical for plasterboard, vehicle bodies, and other thin layered materials), $f_{\text{coh}} = \tau f_{\text{KA}} + (1-\tau) f_{\text{SPM}}$ combines a GGX microfacet specular lobe and a von Mises–Fisher SPM lobe, and $f_{\text{inc}} = \gamma f_{\text{dir}} + (1-\gamma) f_{\text{broad}}$ combines a directional and a Lambertian diffuse lobe. Polarization is tracked through the local $s/p$ basis with complex Jones reflection coefficients per polarization. We reuse mmIR's per-lobe forms unchanged Anonymous [2026]; per-point optimization yields one $\boldsymbol{\theta}_i$ vector per scene point rather than per mesh vertex.

### 3.4 Differentiable optimization

**Learned parameters.** For each point we learn (i) the 6-parameter ITU material $\boldsymbol{\theta}_i$ (shared init, per-point freedom afterward); (ii) the surface-normal quaternion $\mathbf{q}_i$, initialized from the LiDAR normal; and (iii) the 3D position $\mathbf{p}_i$, optimized through the *amplitude path only*. The carrier-phase term $e^{-j\phi_{TRi}}$ in Eq. (2) is detached from the autograd graph for position gradients because the rounded range-bin index $k^\star = \mathrm{round}(r/\Delta r)$ creates a piecewise-constant dependence on $\mathbf{p}_i$: $\partial k^\star/\partial \mathbf{p}_i$ is zero almost everywhere and an unbounded delta at bin boundaries, so phase gradients on position are degenerate. Detaching the carrier phase removes the discontinuity from the backward pass without affecting the forward pass; the carrier phase is still computed from the un-rounded $d_T + d_R$. An L2 anchor $\lambda_{\text{pos}}\|\mathbf{p}_i - \mathbf{p}_i^{(0)}\|^2$ ($\lambda_{\text{pos}}$=100) restricts position drift to $\sim$1 mm per iteration — much smaller than the carrier wavelength ($\lambda \approx 3.9$ mm at 76 GHz) — so geometry remains LiDAR-anchored and the carrier phase remains correct in expectation.

**Loss.** For each training pose $F$ we render the magnitude RA map $|\mathrm{RA}_F^{\text{pred}}|$ via Eq. (2) plus azimuth DFT and compute MSE against the measured $|\mathrm{RA}_F^{\text{GT}}|$, both normalized by their respective $L^2$ norms. The total loss is the mean over training samples; rotation and material drift are unregularized and bounded implicitly by Adam.

**Adaptive density control.** Following 3DGS Kerbl et al. [2023] but adapted to radar physics, we periodically split high-importance points and prune low-importance points to redistribute the fixed $N$ budget. The per-point importance signal is the accumulated amplitude-path position-gradient magnitude $g_i = \sum_F \|\nabla_{\mathbf{p}_i} \mathcal{L}_F\|$ between densify rounds: a high $g_i$ indicates the renderer disagrees with the data at that point, so the point either needs additional capacity (split) or is contributing nothing (prune). To compute $g_i$ without actually moving points, we track the gradient on positions while *excluding* them from the optimizer step, so positions remain on the LiDAR scaffold while the gradient drives the splitting decision. At iters $\{100, 200, 300, 400\}$ we (i) clone the top 5% of points by $g_i$ into Gaussian-jittered children at $\sigma_p = 2$ cm, $\sigma_q = 0.05$ rad, (ii) evict the bottom 5% by $g_i$, and (iii) reset Adam state for the affected slots. The budget $N$=20,000 is exactly preserved.

**Implementation.** Optimization uses Adam ($\beta_1$=0.9, $\beta_2$=0.999) with per-parameter learning rates $\eta_{\text{mat}}$=$10^{-2}$, $\eta_{\text{rot}}$=$5\cdot10^{-3}$, $\eta_{\text{pos}}$=$10^{-5}$, and per-parameter root-mean-square gradient clipping. Each scene's 500-iteration training loop completes in $\sim$3 minutes on a single RTX 4090.

# 4 Experiments

We evaluate 3DPS on six outdoor ColoRadar Kramer et al. [2022] scenes captured with a TI MMWCAS cascaded mmWave radar (12 TX $\times$ 16 RX, 76 GHz carrier, 5.93 cm range resolution) and a co-registered Ouster OS1-64 LiDAR. The scenes span parking lots, vehicle staging areas, and outdoor infrastructure with primarily static structure (walls, parked vehicles, traffic infrastructure).

## 4.1 Experimental setup

**Scene splits.** For each scene we select 9 adjacent cascaded radar frames at 5 Hz (200 ms inter-frame spacing) and use only the first chirp loop of each frame. The middle frame $F$ is held out as the novel-view test frame and the eight bracketing frames $\{F\pm1,\ldots,\pm4\}$ form the training set, giving 8 training RA maps and 1 held-out test RA map per scene. The test pose is reconstructed via per-loop interpolation between the held-out frame's neighbours rather than read from the test frame's own ground-truth alignment, so the test pose is itself NVS-inferred.

**Baselines.** We benchmark against three published radar NVS methods using public reference implementations: **DART** Huang et al. [2024] (RDA neural field, run on cascade data with a custom 86-element azimuth gain pattern at $N_d$=16 chirps per frame), **Radar Fields** Borts et al. [2024] (hash-grid + MLP, RA magnitude), and **RadarSplat** Kung et al. [2025] (3DGS adapted to RA rendering with opaque per-Gaussian features). All three render power-only RA maps, so comparisons are performed on $|\mathrm{RA}|$ even though 3DPS produces complex range profiles.

**Metrics.** Following Kramer et al. Kramer et al. [2022] and mmIR Anonymous [2026], we report Pearson correlation (Corr), peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and root-mean-square error (RMSE), all computed on min-max-normalized $399\times399$ Cartesian $|\mathrm{RA}|$ images via a shared metric harness used by all four methods. Each pixel is bilinearly sampled from the polar render. Range bins 15–110 are retained and the central wedge is masked to the radar's actual azimuth FOV (matching the standard ColoRadar evaluation).

**Implementation.** 3DPS uses 20,000 oriented points per scene with per-point ITU material vectors and quaternion-encoded normals, trained for 500 Adam iterations (Sec. 3.4). Each scene completes in $\sim$3.2 minutes on a single NVIDIA RTX 4090 (range 2.3–3.7 min across the six scenes); measured baseline wall-clocks on the same hardware are 0.4 min/scene (DART), 1.6 min/scene (Radar Fields), and 19.7 min/scene (RadarSplat). Under the same hardware budget mmIR Anonymous [2026] would project to $\sim$160 min per scene (8 viewpoints $\times$ $\sim$20 min/viewpoint sequential), a $\sim$50$\times$ per-scene speedup. Per-scene wall-clock and peak GPU memory are reported in the supplement (Sec. D).

## 4.2 Quantitative results

Table 2 reports the cross-method, cross-product results: mean over the six ColoRadar scenes, separately for training views (8 per scene) and the held-out novel-view test frame. The three baselines render power-only $|\mathrm{RA}|$ by construction and have nothing to report on CRP, ADC, or complex metrics — those columns are marked '–'. Per-scene breakdowns and the full discussion of CRP/ADC eval methodology (range Hann, azimuth Hann, near-field bin masking, per-VA $\alpha[v]$ + per-range $\beta[r]$ phase calibration removal) are in supplement Sec. A and B.

**Magnitude on $|\mathrm{RA}|$.** 3DPS achieves **0.812 train / 0.587 test** mean Pearson correlation on cart $|\mathrm{RA}|$, beating the next-best baseline by $2.2\times$ (train) and $1.8\times$ (test). PSNR / SSIM / RMSE all rank 3DPS first on the mean, and on every per-scene metric where reported (supplement Tables 3 and 4). The numerical ranking matches the structural mismatches identified in Sec. 2: RadarSplat's opaque per-Gaussian features cannot share information with the radar's antenna pattern (capping at 0.37 / 0.33); Radar Fields' hash-grid+MLP supervised post-FFT, lacking a virtual-array model, fails to discover scene support from sparse RA alone (0.13 / 0.12, data-starved); DART, designed for $N_d$=256 chirps and operating at the cascade's $N_d$=16, has insufficient Doppler resolution (0.03 / 0.08).

Table 2: **Cross-method, cross-product results on six ColoRadar scenes (mean across scenes).** Magnitude: Pearson Corr / PSNR / SSIM / MSE on min-max-normalized $|\cdot|$. Complex: $|\rho|$, complex PSNR (CRP), and magnitude-weighted phase RMSE $\sigma_\phi$ (ADC; degrees; $104° =$ random) after absolute phase calibration. $|\rho|$ is reported once (CRP $\equiv$ ADC by Parseval). Baselines emit magnitude-only RA, so CRP/ADC/Complex columns are marked '–'. Per-scene breakdowns in supplement Tables 3–6.

| | **RA cart** | | | | **CRP** | | | | | **ADC** | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Magnitude** | | | | **Magnitude** | | | **Complex** | | **Env.** | **Complex** |
| Method | Corr | PSNR | SSIM | RMSE | Corr | PSNR | SSIM | $\lvert\rho\rvert$ | $\mathrm{PSNR}_{\mathbb{C}}$ | $\rho_{\lvert\cdot\rvert}$ | $\sigma_\phi$ |
| **Train Mean** | | | | | | | | | | | |
| DART Huang et al. [2024] | 0.033 | 23.8 | 0.537 | 0.0679 | – | – | – | – | – | – | – |
| Radar Fields Borts et al. [2024] | 0.129 | 24.5 | 0.478 | 0.0612 | – | – | – | – | – | – | – |
| RadarSplat Kung et al. [2025] | 0.371 | 25.5 | 0.532 | 0.0555 | – | – | – | – | – | – | – |
| **3DPS (Ours)** | **0.812** | **31.9** | **0.819** | **0.0272** | **0.701** | **27.9** | **0.721** | **0.430** | **24.7** | **0.689** | **65.8** |
| **Test Mean** | | | | | | | | | | | |
| DART Huang et al. [2024] | 0.081 | 23.4 | 0.433 | 0.0690 | – | – | – | – | – | – | – |
| Radar Fields Borts et al. [2024] | 0.120 | 28.0 | 0.495 | 0.0402 | – | – | – | – | – | – | – |
| RadarSplat Kung et al. [2025] | 0.333 | 25.3 | 0.522 | 0.0570 | – | – | – | – | – | – | – |
| **3DPS (Ours)** | **0.587** | **28.8** | **0.734** | **0.0369** | **0.603** | **24.7** | **0.650** | **0.255** | **21.1** | **0.613** | **82.5** |

**Magnitude on CRP and ADC.** 3DPS' native complex range profile gives mean train / test CRP Corr **0.701 / 0.603** — directly comparable to the $|\mathrm{RA}|$ Corr in the same table, with the expected mild attenuation due to per-virtual-antenna evaluation being inherently stricter than azimuth-integrated RA (the azimuth FFT in $|\mathrm{RA}|$ provides $\sim\sqrt{86}$ coherent gain at target DOAs that per-(v,r) $|\mathrm{CRP}|$ does not). ADC envelope $\rho_{|\cdot|}$ reaches **0.689 / 0.613**. ADC drops PSNR/SSIM (image-domain, inappropriate for raw I/Q time-series). *No baseline produces these outputs at all,* so the column is filled with '–' for the three baselines.

**Complex output.** The renderer's per-(TX,RX) phasor is constrained only indirectly by the magnitude-only $|\mathrm{RA}|$ loss; the per-RA-bin phase is unconstrained by design ($\sim 22{,}000$ phase DOFs). After a joint per-VA $\alpha[v]$ + per-range $\beta[r]$ least-squares phase calibration removal (342 DOFs $\approx 1.5\%$ of phase content, physically motivated by per-channel RF drift and ADC sample-zero timing), 3DPS reaches mean train / test $|\rho|$ of **0.430 / 0.255** and magnitude-weighted phase RMSE $\sigma_\phi$ of **65.8**° / **82.5**° (vs. 104° for random phase). *Again, no baseline produces complex output, so these columns are inapplicable to the three optical-derivative methods.*

**Qualitative comparison.** Figure 3 shows held-out test $|\mathrm{RA}|$ renders for all four methods on the six scenes, sorted left-to-right by 3DPS test CC (descending). 3DPS preserves dominant scene structure (walls, parked vehicles, support columns) with consistent range and azimuth localization across all scenes. Per-scene side-by-side comparisons of every training frame plus the held-out test frame are in Sec. C.1. CRP and ADC qualitative heatmaps are in Sec. C.2; wall-clock and peak GPU memory in Sec. D; extended limitations and planned future work (ablations, per-point material visualizations) in Sec. E.

## 5 Conclusion

We presented 3DPS, a differentiable physically based point renderer for FMCW mmWave radar that synthesizes complex range profiles from novel sensor poses. The central design choice — replacing optical-radiance primitives with oriented 3D points carrying ITU-R P.2040 material parameters, evaluated by closed-form BSDFs and splatted through precomputed sensor PSFs — eliminates the Monte Carlo sampling and post-FFT supervision that constrain prior radar-NVS methods, yielding a renderer that is simultaneously physics-faithful, multi-viewpoint-tractable, and complex-valued. On six outdoor ColoRadar scenes, 3DPS achieves **0.587** mean Pearson correlation on held-out novel-view $|\mathrm{RA}|$ — $1.8\times$, $4.9\times$, and $7.2\times$ the strongest published baselines (RadarSplat, Radar Fields, DART) — with physically interpretable per-point materials, in $\sim$3 minutes per scene over 8 training viewpoints ($\sim 50\times$ faster than mmIR's per-scene MC ray tracing cost of $\sim$160 min).

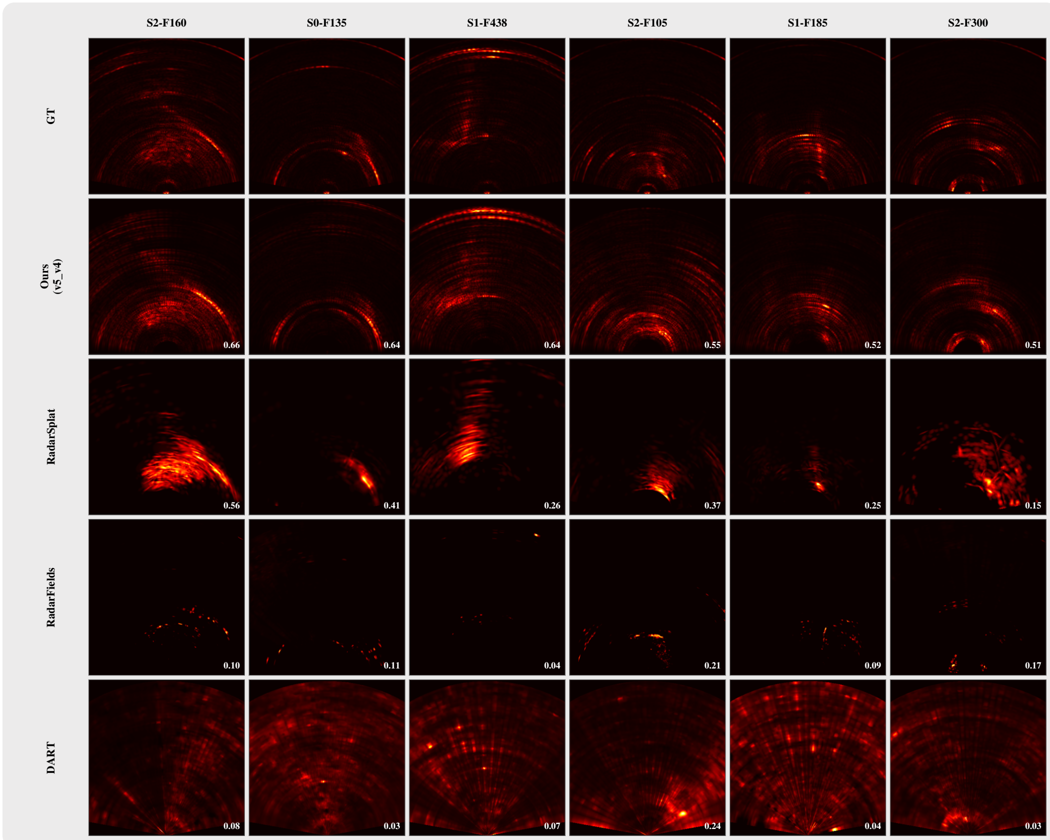


Figure 3: **Held-out novel-view $|\mathrm{RA}|$ comparison.** Columns: six ColoRadar scenes (sorted by 3DPS test CC, descending). Rows: GT, 3DPS (Ours), RadarSplat, Radar Fields, DART. Per-frame test CC overlaid.

**Limitations and future work.** The current renderer evaluates single-bounce paths only, sufficient for the static outdoor scenes we target but not for indoor multipath-dominated environments where mmIR-style multi-bounce path tracing remains necessary. The geometric scaffold relies on a co-registered LiDAR cloud; relaxing this to LiDAR-free initialization is an open direction for purely radar-driven NVS. We supervise on $|\mathrm{RA}|$ magnitude for fair comparison with the three baselines, but a phase-aware loss exploiting the renderer's complex output should narrow the train–test gap further. Extension to dynamic scenes (range-Doppler-coherent rendering across moving objects), to downstream perception tasks (radar-trained detectors using NVS-augmented data), and to broader radar geometries beyond cascaded MIMO are natural next steps.

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Table 3: **Per-scene held-out test $|\mathrm{RA}|$ metrics on six ColoRadar scenes.** Best per scene in **bold**. Mean row anchors the corresponding test row of main paper Table 2.

| | RA Corr ↑ | | | | RA PSNR ↑ | | | | RA SSIM ↑ | | | | RA RMSE ↓ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Scene | Ours | RSplat | RFields | DART | Ours | RSplat | RFields | DART | Ours | RSplat | RFields | DART | Ours | RSplat | RFields | DART |
| S0 F135 | **0.644** | 0.411 | 0.107 | 0.028 | **31.3** | 27.8 | 29.1 | 23.0 | **0.839** | 0.557 | 0.617 | 0.355 | **0.0273** | 0.0405 | 0.0349 | 0.0709 |
| S1 F185 | **0.519** | 0.248 | 0.085 | 0.043 | **29.5** | 28.3 | 28.3 | 20.9 | **0.780** | 0.615 | 0.579 | 0.290 | **0.0333** | 0.0385 | 0.0385 | 0.0898 |
| S1 F438 | **0.642** | 0.260 | 0.040 | 0.066 | **27.1** | 23.4 | 27.1 | 23.5 | **0.663** | 0.513 | 0.434 | 0.439 | **0.0442** | 0.0677 | 0.0444 | 0.0672 |
| S2 F105 | **0.553** | 0.366 | 0.210 | 0.237 | 27.6 | 26.6 | **28.0** | 22.4 | **0.683** | 0.521 | 0.438 | 0.505 | 0.0415 | 0.0469 | **0.0399** | 0.0755 |
| S2 F160 | **0.657** | 0.564 | 0.103 | 0.079 | **27.7** | 21.2 | 26.3 | 26.3 | **0.653** | 0.392 | 0.315 | 0.607 | **0.0413** | 0.0873 | 0.0483 | 0.0484 |
| S2 F300 | **0.508** | 0.147 | 0.172 | 0.034 | **29.4** | 24.3 | 29.0 | 24.1 | **0.785** | 0.532 | 0.590 | 0.404 | **0.0340** | 0.0613 | 0.0353 | 0.0621 |
| **Mean** | **0.587** | 0.333 | 0.120 | 0.081 | **28.8** | 25.3 | 28.0 | 23.4 | **0.734** | 0.522 | 0.495 | 0.433 | **0.0369** | 0.0570 | 0.0402 | 0.0690 |

Table 4: **Per-scene training-view $|\mathrm{RA}|$ metrics on six ColoRadar scenes.** Each cell is the mean over the 8 training frames per scene. Same metric harness as Table 3. Best per scene in **bold**. Mean row anchors the corresponding train row of main paper Table 2.

| | RA Corr ↑ | | | | RA PSNR ↑ | | | | RA SSIM ↑ | | | | RA RMSE ↓ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Scene | Ours | RSplat | RFields | DART | Ours | RSplat | RFields | DART | Ours | RSplat | RFields | DART | Ours | RSplat | RFields | DART |
| S0 F135 | **0.822** | 0.346 | 0.177 | -0.030 | **32.0** | 26.2 | 25.2 | 22.9 | **0.837** | 0.545 | 0.521 | 0.483 | **0.0261** | 0.0497 | 0.0551 | 0.0733 |
| S1 F185 | **0.836** | 0.381 | 0.127 | 0.015 | **32.9** | 29.2 | 25.0 | 25.2 | **0.819** | 0.642 | 0.594 | 0.466 | **0.0266** | 0.0363 | 0.0571 | 0.0561 |
| S1 F438 | **0.848** | 0.264 | 0.056 | 0.034 | **32.6** | 24.0 | 22.8 | 25.2 | **0.852** | 0.515 | 0.416 | 0.559 | **0.0241** | 0.0650 | 0.0728 | 0.0572 |
| S2 F105 | **0.804** | 0.430 | 0.163 | 0.156 | **31.3** | 26.0 | 24.2 | 24.3 | **0.815** | 0.495 | 0.408 | 0.691 | **0.0289** | 0.0506 | 0.0617 | 0.0622 |
| S2 F160 | **0.784** | 0.589 | 0.164 | -0.041 | **29.7** | 22.9 | 23.0 | 20.0 | **0.743** | 0.390 | 0.294 | 0.530 | **0.0338** | 0.0719 | 0.0715 | 0.1009 |
| S2 F300 | **0.774** | 0.218 | 0.086 | 0.062 | **32.9** | 24.8 | 26.5 | 25.4 | **0.847** | 0.602 | 0.636 | 0.493 | **0.0236** | 0.0595 | 0.0490 | 0.0576 |
| **Mean** | **0.812** | 0.371 | 0.129 | 0.033 | **31.9** | 25.5 | 24.5 | 23.8 | **0.819** | 0.532 | 0.478 | 0.537 | **0.0272** | 0.0555 | 0.0612 | 0.0679 |

The supplement provides the per-scene quantitative evidence and qualitative comparisons that anchor the main paper's claims, the methodology details for the CRP and ADC evaluation introduced in Sec. 4.2, implementation details (wall-clock, hardware), and a more detailed discussion of limitations and future work.

# A Per-scene $|\mathrm{RA}|$ breakdowns

The main paper Table 2 reports the 6-scene mean for each method on training views (8 views per scene) and the held-out test view. Tables 3 and 4 below give the per-scene breakdown that anchors those means, retaining the canonical $|\mathrm{RA}|$ metric set (Corr, PSNR, SSIM, RMSE on min-max-normalized $399 \times 399$ Cartesian images via the shared `compute_cartesian_ra_metrics` harness). The Mean row of each table matches the corresponding 3DPS row in Table 2 by construction.

# B CRP and ADC evaluation

Because 3DPS produces complex range profiles, the same optimized scene yields range–azimuth and raw ADC outputs by applying the standard azimuth FFT and inverse range FFT to the per-(TX, RX) range-profile output of Eq. (2) — with no retraining. The aggregate fidelity numbers were reported in Table 2 of the main paper. This section provides the eval pipeline conventions and the per-scene CRP/ADC tables that anchor the 3DPS row of Table 2.

## B.1 Pipeline conventions

CRP and ADC are evaluated in the trainer's loss domain to enable direct comparison with the published $|\mathrm{RA}|$ numbers. The GT side applies range Hann + range FFT to the raw ADC; both GT and predicted CRPs then receive the azimuth Hann window on the virtual-array axis (matching `adc_to_ra_complex` and `range_profile_to_ra` in the trainer). Range bins $[0{:}15]$ are zeroed in both signals to reject TX–RX coupling. ADC is the inverse range FFT of the resulting CRP. Magnitude metrics are invariant to phase correction; complex metrics depend on it. A round-trip FFT/IFFT sanity test (no Hann, no zero-pad) is exact to $5 \times 10^{-16}$ relative error on every scene.

Table 5: **Per-scene CRP and ADC fidelity (training-view).** Same metric set and column convention as main paper Table 2. Mean row anchors the 3DPS (Ours) train row of that table.

| | **CRP** | | | | | | **ADC** | |
|---|---|---|---|---|---|---|---|---|
| | | **Magnitude** | | | **Complex** | | **Env.** | **Complex** |
| Scene | Corr | PSNR | SSIM | MSE | $\lvert\rho\rvert$ | $\text{PSNR}_{\mathbb{C}}$ | $\rho_{\lvert\cdot\rvert}$ | $\sigma_\phi$ |
| S0 F135 | 0.677 | 28.1 | 0.697 | 0.0016 | 0.448 | 25.7 | 0.665 | 62.7 |
| S1 F185 | 0.741 | 26.5 | 0.753 | 0.0024 | 0.418 | 22.5 | 0.674 | 68.6 |
| S1 F438 | 0.718 | 29.6 | 0.742 | 0.0012 | 0.500 | 26.6 | 0.725 | 60.3 |
| S2 F105 | 0.690 | 27.3 | 0.707 | 0.0022 | 0.370 | 23.9 | 0.666 | 70.8 |
| S2 F160 | 0.671 | 26.4 | 0.627 | 0.0028 | 0.369 | 22.9 | 0.681 | 71.2 |
| S2 F300 | 0.709 | 29.6 | 0.798 | 0.0014 | 0.475 | 26.8 | 0.719 | 61.4 |
| **Mean** | **0.701** | **27.9** | **0.721** | **0.0020** | **0.430** | **24.7** | **0.689** | **65.8** |

Table 6: **Per-scene CRP and ADC fidelity (held-out test).** Same metric set and column convention as main paper Table 2. Mean row anchors the 3DPS (Ours) test row of that table.

| | **CRP** | | | | | | **ADC** | |
|---|---|---|---|---|---|---|---|---|
| | | **Magnitude** | | | **Complex** | | **Env.** | **Complex** |
| Scene | Corr | PSNR | SSIM | MSE | $\lvert\rho\rvert$ | $\text{PSNR}_{\mathbb{C}}$ | $\rho_{\lvert\cdot\rvert}$ | $\sigma_\phi$ |
| S0 F135 | 0.626 | 26.2 | 0.714 | 0.0024 | 0.314 | 23.2 | 0.667 | 78.9 |
| S1 F185 | 0.594 | 26.0 | 0.705 | 0.0025 | 0.227 | 22.6 | 0.612 | 84.6 |
| S1 F438 | 0.554 | 25.0 | 0.632 | 0.0032 | 0.278 | 21.8 | 0.586 | 81.4 |
| S2 F105 | 0.611 | 23.2 | 0.639 | 0.0048 | 0.221 | 19.1 | 0.590 | 84.4 |
| S2 F160 | 0.615 | 23.6 | 0.536 | 0.0044 | 0.203 | 19.3 | 0.588 | 86.3 |
| S2 F300 | 0.619 | 24.2 | 0.671 | 0.0038 | 0.288 | 20.9 | 0.636 | 79.7 |
| **Mean** | **0.603** | **24.7** | **0.650** | **0.0035** | **0.255** | **21.1** | **0.613** | **82.5** |

### B.2 Per-scene CRP and ADC fidelity

Tables 5 and 6 give the per-scene CRP/ADC fidelity that anchors the 3DPS row of Table 2. For each scene, the value is the 8-train-frame mean (train table) or the held-out test frame value (test table); the Mean row at the bottom of each is the average across the six scenes and matches the corresponding 3DPS Train Mean / Test Mean row of Table 2 by construction. Same metric set and column layout as Table 2.

## C Qualitative comparisons

### C.1 Per-scene $|\mathrm{RA}|$ comparison across methods

Figures 4–9 show per-scene side-by-side $|\mathrm{RA}|$ renders for all four methods on every training frame, plus the held-out test frame ($F$, last column with the red header). Rows: GT, 3DPS (Ours), RadarSplat, Radar Fields, DART (cascaded). Per-cell train/test CC overlaid in white. These figures complement the per-scene tables in Sec. A and confirm visually that 3DPS preserves wall and vehicle structure across all eight training frames as well as the held-out test frame, while the three baselines exhibit the failure modes discussed in Sec. 4.2.

### C.2 CRP and ADC heatmaps and I/Q traces

Figures 10–12 show the qualitative side-by-side comparison of $|\mathrm{CRP}|$ and $|\mathrm{ADC}|$ between GT and the VR-corrected prediction across all six scenes (held-out test frames). Each panel is normalized identically to its $|\mathrm{RA}|$ counterpart in the main paper, so the visual structure is directly comparable.

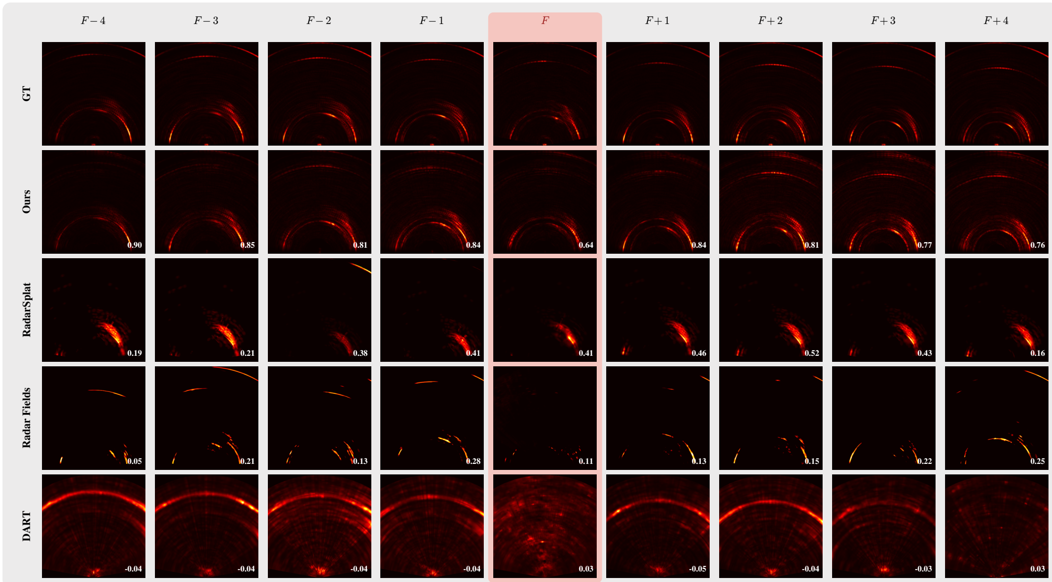


Figure 4: **Per-frame** $|\mathrm{RA}|$ **comparison, scene S0 F135.** Cols: 8 train frames ($F\pm1\ldots\pm4$) + held-out test frame $F$ (red header). Rows: GT, 3DPS, RadarSplat, Radar Fields, DART. Per-frame CC overlaid.

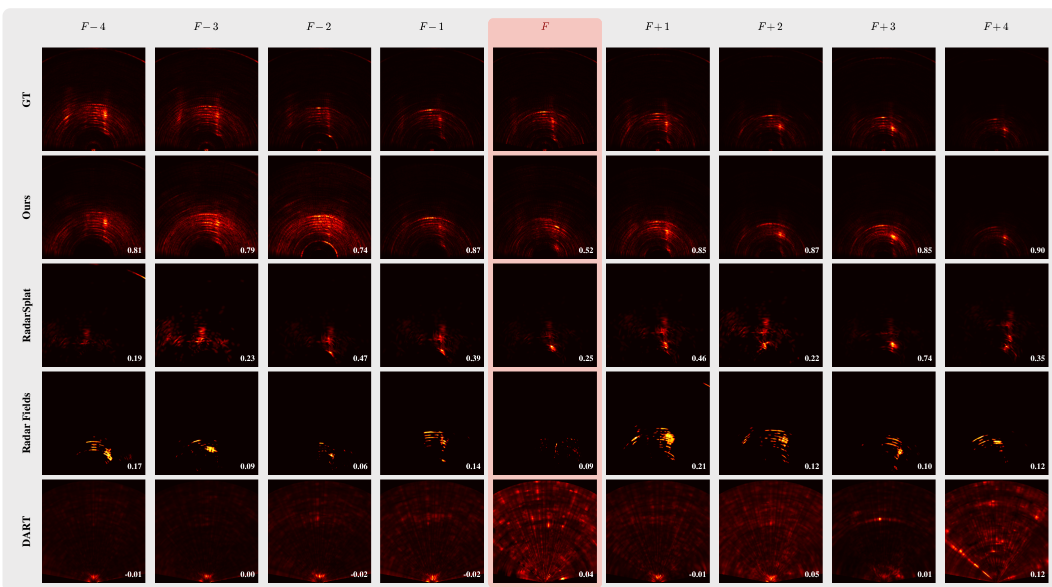


Figure 5: **Per-frame** $|\mathrm{RA}|$ **comparison, scene S1 F185.**

# D Wall-clock and peak GPU memory

Table 7 reports per-scene wall-clock and peak GPU memory measured on a single NVIDIA RTX 4090 (24 GB) for 3DPS and the three baselines, on the same 6-scene ColoRadar benchmark and the same 9-frame split (8 training viewpoints, 1 held-out test frame). All measurements include forward+backward+optimizer+I/O. mmIR's projected cost is $8\times\sim$20 min/viewpoint $=\sim$160 min for the 8-viewpoint optimization that NVS demands; we did not run mmIR end-to-end on this benchmark because its per-viewpoint MC cost makes the multi-viewpoint task intractable in the optimization wall-clock window.

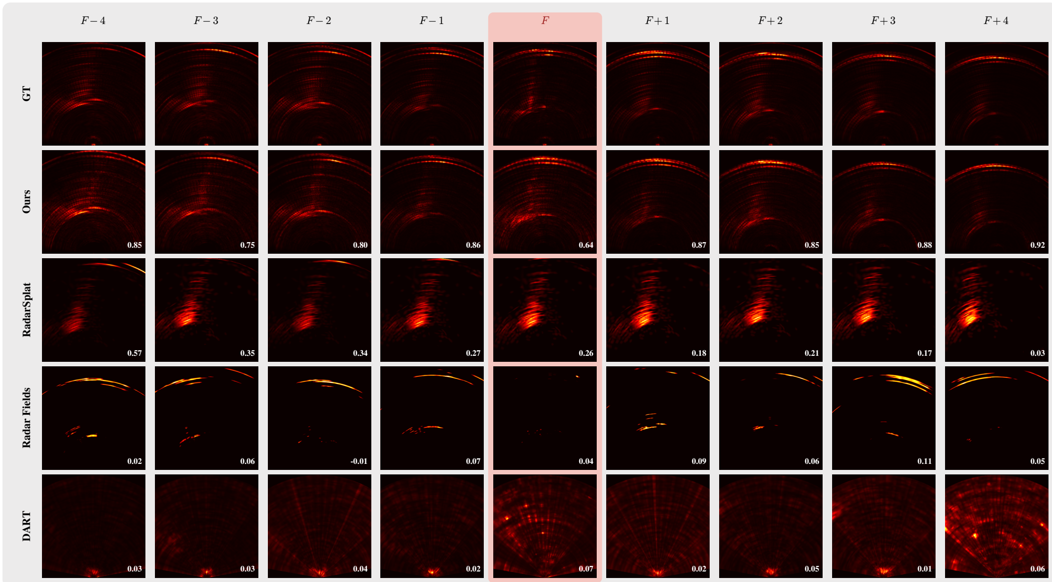


Figure 6: **Per-frame** $|\mathrm{RA}|$ **comparison, scene S1 F438.**

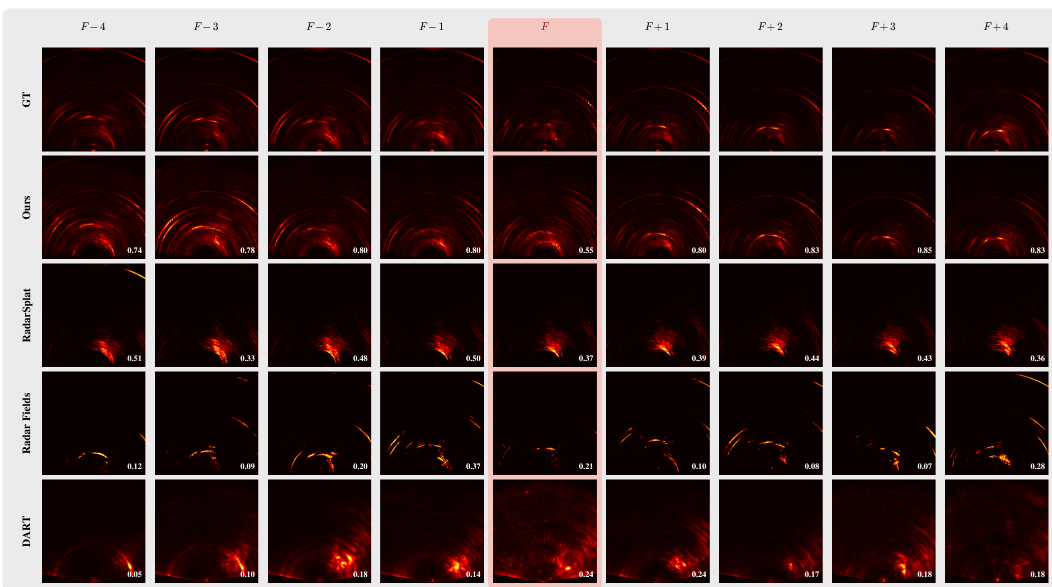


Figure 7: **Per-frame** $|\mathrm{RA}|$ **comparison, scene S2 F105.**

# E Limitations and future work

The conclusion (Sec. 5) summarizes the main limitations briefly. This section expands on each and identifies the corresponding planned investigations.

**Single-bounce path tracing.** The renderer evaluates a single TX→scatterer→RX path per oriented point. For static outdoor scenes dominated by direct surface returns (the regime of our ColoRadar benchmark) this is sufficient, and the headline 0.587 mean test $|\mathrm{RA}|$ Pearson reflects that. For indoor multipath-dominated environments where second- and third-order reflections carry comparable energy to direct returns, multi-bounce path tracing in the style of mmIR Anonymous [2026] would remain necessary; integrating the closed-form BSDF + PSF splatting recipe into a multi-bounce framework is open.

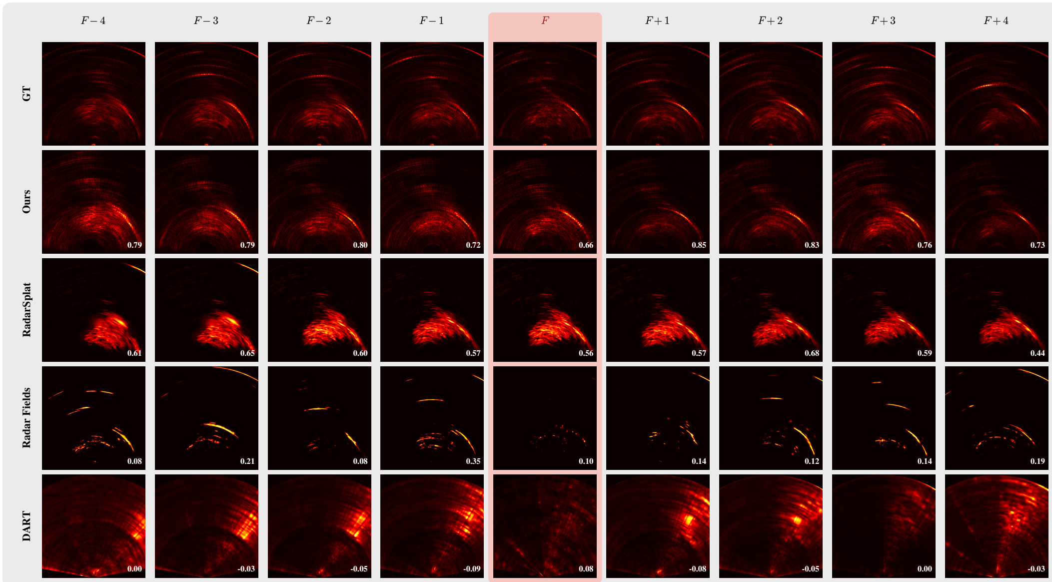


Figure 8: **Per-frame $|\mathrm{RA}|$ comparison, scene S2 F160.**

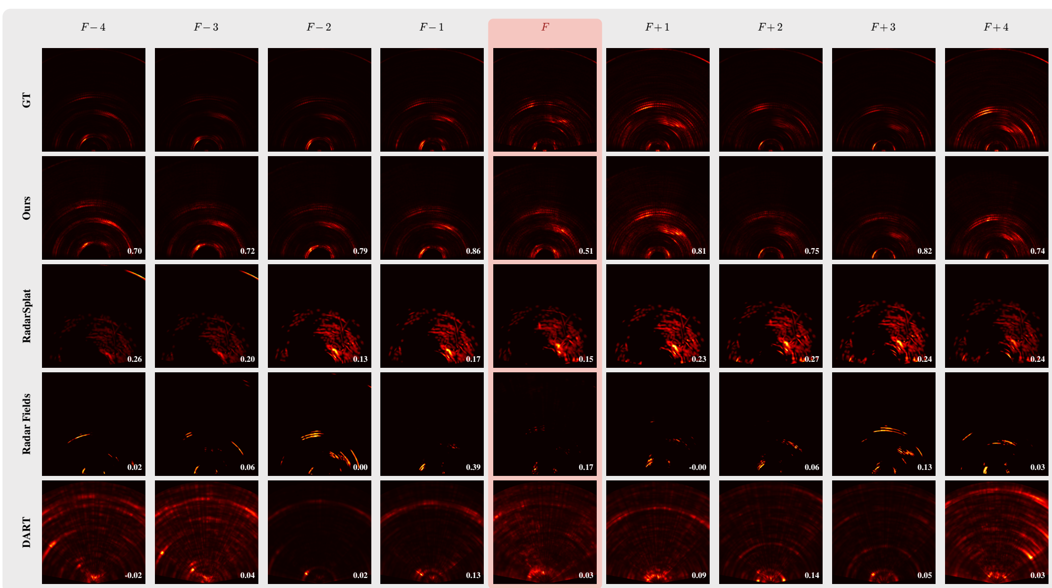


Figure 9: **Per-frame $|\mathrm{RA}|$ comparison, scene S2 F300.**

**Magnitude-only loss leaves per-RA-bin phase unconstrained.** 3DPS supervises on $|\mathrm{RA}|$ for fair comparison with the three magnitude-only baselines, leaving $\sim 22{,}000$ per-RA-bin phase DOFs unconstrained by the loss. The rank-1 (per-VA $\alpha[v]$ + per-range $\beta[r]$) phase calibration removal we apply at evaluation time absorbs known per-channel RF drift and ADC sample-zero timing (342 DOFs, $\sim$1.5% of phase content); a tighter phase fit would require adding a complex (Re/Im) loss term during training. A natural future extension is a two-stage training recipe (first stage: magnitude loss to fit geometry and material; second stage: bounded position refinement under a complex loss to recover phase) that locks in the magnitude performance and adds phase fidelity on top.

**LiDAR-dependent initialization.** The point cloud that seeds 3DPS comes from a co-registered LiDAR scan. This is a strong prior that simplifies geometric initialization and accelerates convergence to a reasonable scene fit. Pure radar-driven NVS — removing the LiDAR dependence — is an important direction for radar-only deployments and an open challenge for our framework.

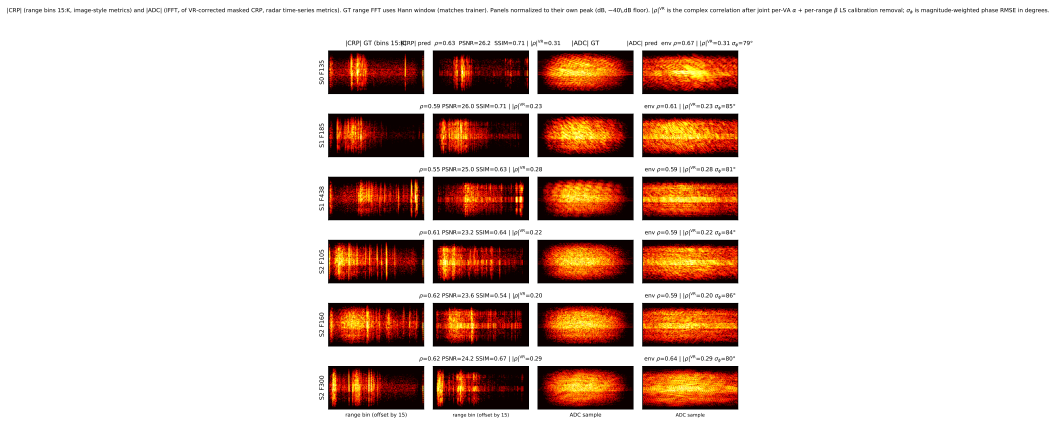


Figure 10: $|\mathrm{CRP}|$ **and** $|\mathrm{ADC}|$ **heatmaps on held-out test frames (dB scale).** Rows: scenes. Cols: GT and predicted $|\mathrm{CRP}|$ and $|\mathrm{ADC}|$. CRP cropped to range bins $[15{:}256]$ (the metric domain); ADC is IFFT$_r$ of the calibration-corrected, near-field-masked CRP. Each panel peak-normalized to its own max with a $-40\,$dB floor.

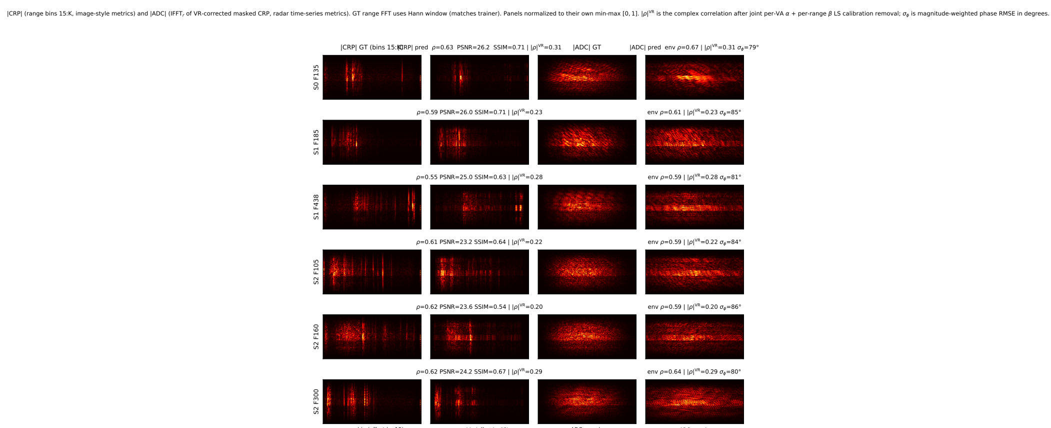


Figure 11: **Linear-scale equivalent of Fig. 10.** Panels min-max normalized to $[0, 1]$ — the metric domain.

**Dynamic scenes and downstream perception.** Extending range-Doppler-coherent rendering to scenes with moving targets, and validating NVS-augmented training data on radar-trained perception tasks (object detection, semantic segmentation), are natural next steps that the product-agnostic complex-output property of 3DPS enables but does not directly address.

# F Ablations

**[Placeholder — to be reported in camera-ready / extended version.]** Planned ablations on the 6-scene ColoRadar benchmark: (i) point count $N \in \{2k, 5k, 10k, 20k, 50k\}$, (ii) adaptive density control on/off, (iii) carrier-phase detach for position gradients, (iv) the four-stage LiDAR initialization recipe (each stage individually disabled). Default 3DPS configuration is the bolded top row.

# G Per-point material visualizations

**[Placeholder — to be reported in camera-ready / extended version.]** Planned: render the optimized point clouds colored by each ITU-R P.2040 parameter ($\varepsilon_r'$, $\varepsilon_r''$, $\sigma_h$, $l_c$, $\tau$, $d$), and report per-cluster material distributions partitioned by LiDAR class label (walls, vehicles, ground). Supports the "physically interpretable per-point materials" claim of Sec. 1.

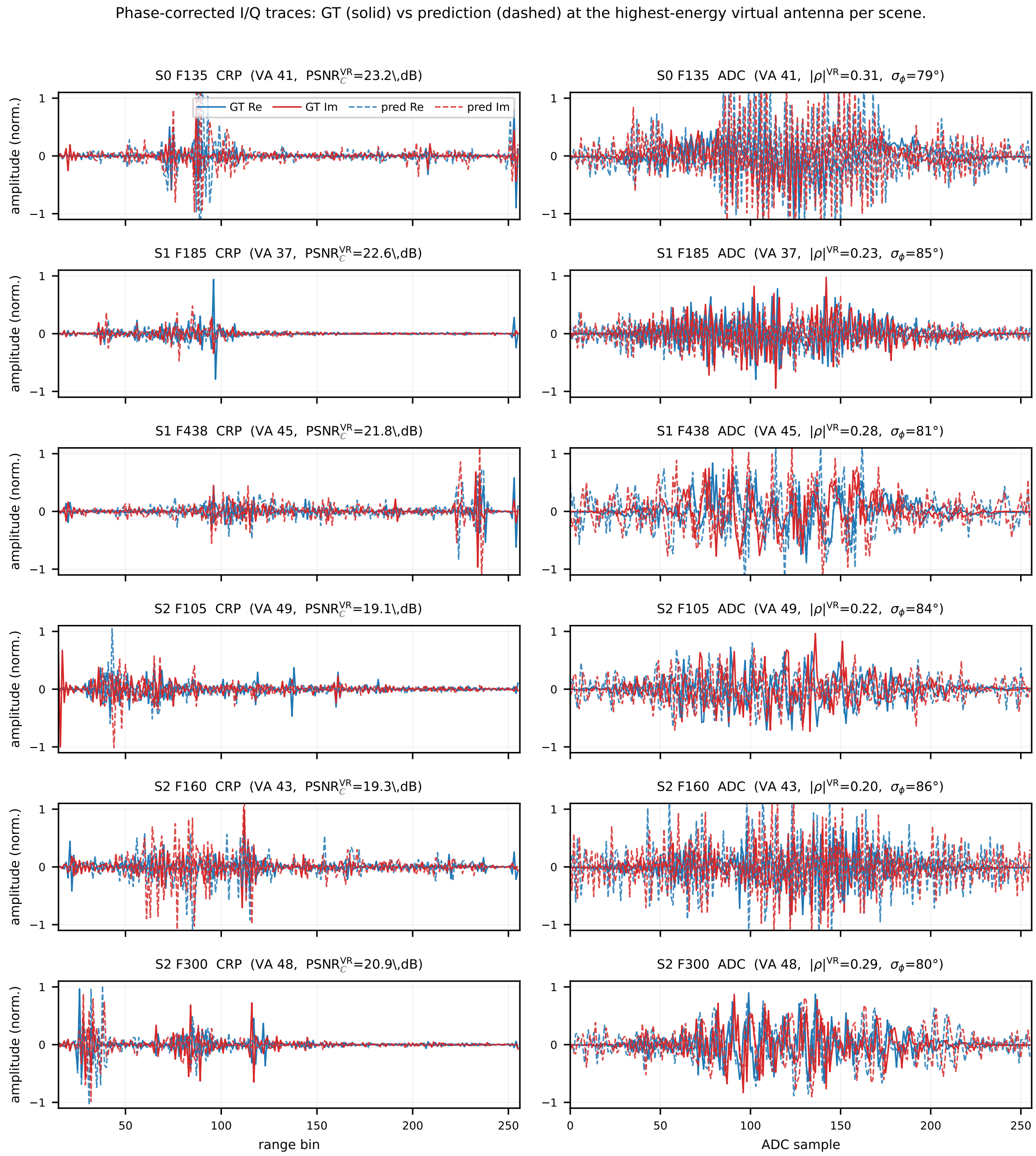


Figure 12: **Calibration-corrected I/Q traces** for the highest-energy VA per scene. Solid: GT real (blue) and imaginary (red); dashed: prediction. Left: CRP across range bins $[15{:}256]$. Right: ADC across 256 fast-time samples. Per-panel $|\rho|$ and $\sigma_\phi$ in title.

Table 7: **Wall-clock and peak GPU memory per scene** on the 6-scene benchmark, single RTX 4090. Wall-clock: mean (min) across scenes; range is per-scene min–max. mmIR is projected from its per-viewpoint cost.

| **Method** | **Mean wall-clock (min)** | **Per-scene range (min)** | **Peak GPU mem (GB)** |
|---|---|---|---|
| DART (cascaded) Huang et al. [2024] | 0.43 | 0.4–0.5 | not reported |
| Radar Fields Borts et al. [2024] | 1.63 | 1.6–1.7 | 2.4 |
| RadarSplat Kung et al. [2025] | 19.74 | 14.2–21.4 | 14.2 |
| mmIR Anonymous [2026] (projected, 8 viewpoints) | $\sim$160 | — | — |
| **3DPS (Ours)** | **3.17** | 2.3–3.7 | — |

Table 8: **[Placeholder]** Ablation sweep on the 6-scene benchmark. Mean held-out test Corr, mean training Corr, and mean wall-clock per scene on a single RTX 4090.

| Configuration | Test Corr ↑ | Train Corr ↑ | Wall-clock (min) |
|---|---|---|---|
| **3DPS default** ($N$=20k, ADC on, phase-detach on, full init) | — | — | — |
| $N$= 2,000 | — | — | — |
| $N$= 5,000 | — | — | — |
| $N$=10,000 | — | — | — |
| $N$=50,000 | — | — | — |
| no adaptive density control | — | — | — |
| no phase-detach (position grad.) | — | — | — |
| no cosine-weighted resample | — | — | — |
| no FPS reduction (random sub-sample) | — | — | — |